Practical No.7: Implement program to identify the most important features that contribute to the model accuracy

1. Implement feature engineering techniques to prepare data for machine learning models.

```
import pandas as pd
import numpy as np
from sklearn.preprocessing import StandardScaler
```

```
data = {
        "Age":[22,25,47,52,46,56,55,60,62,30,28,35,40,48,50],

        "Salary":[25000,30000,52000,58000,54000,60000,62000,65000,70000,35000,32000,42000,48000,55000,57000],

        "Gender":["Male","Female","Female","Male","Male","Female","Male","Female","Male","Female","Male",
                  "Female","Male","Female","Male"],

        "Department":["IT","HR","Finance","IT","Finance","HR","IT","Finance","IT","HR","IT","Finance",
                      "HR","IT","Finance"],

        "Purchased":[0,0,1,1,1,1,1,1,1,0,0,0,1,1,1]
}
df = pd.DataFrame(data)
print(df)

print(df.isnull().sum())
```

```
      Age  Salary  Gender Department  Purchased
  0    22   25000    Male         IT          0
  1    25   30000  Female         HR          0
  2    47   52000  Female    Finance          1
  3    52   58000    Male         IT          1
  4    46   54000    Male    Finance          1
  5    56   60000  Female         HR          1
  6    55   62000    Male         IT          1
  7    60   65000  Female    Finance          1
  8    62   70000    Male         IT          1
  9    30   35000  Female         HR          0
  10   28   32000    Male         IT          0
  11   35   42000  Female    Finance          0
  12   40   48000    Male         HR          1
  13   48   55000  Female         IT          1
  14   50   57000    Male    Finance          1
  Age           0
  Salary        0
  Gender        0
  Department    0
  Purchased     0
  dtype: int64
```

```
df_encoded = df.copy()
df_encoded["Gender"]= df_encoded["Gender"].map({"Male":1, "Female":0})
print(df_encoded["Gender"])
```

```
0     1
1     0
2     0
3     1
4     1
5     0
6     1
7     0
8     1
9     0
10    1
11    0
12    1
13    0
14    1
Name: Gender, dtype: int64
```

```python
df_encoded = pd.get_dummies(df_encoded,columns=["Department"],drop_first=True)
```

```python
scaler = StandardScaler()
df_encoded[["Age","Salary"]]= scaler.fit_transform(df_encoded[["Age","Salary"]])
print(df_encoded)
```

```
         Age    Salary  Gender  Purchased  Department_HR  Department_IT
0  -1.730131 -1.849769       1          0          False           True
1  -1.491309 -1.474816       0          0           True          False
2   0.260050  0.174978       0          1          False          False
3   0.658087  0.624922       1          1          False           True
4   0.180443  0.324959       1          1          False          False
5   0.976516  0.774903       0          1           True          False
6   0.896908  0.924884       1          1          False           True
7   1.294945  1.149856       0          1          False          False
8   1.454159  1.524809       1          1          False           True
9  -1.093273 -1.099863       0          0           True          False
10 -1.252487 -1.324834       1          0          False           True
11 -0.695237 -0.574928       0          0          False          False
12 -0.297200 -0.124984       1          1           True          False
13  0.339658  0.399950       0          1          False           True
14  0.498872  0.549931       1          1          False          False
```

2. Write a program to identify the most important features that contributes to the model accuracy.

```
import pandas as pd
import numpy as np

from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score,classification_report
```

```
data = {
    "Study_Hours":[2,4,6,8,10,1,3,5,7,9,4,6,8,2,5],
    "Attendance":[60,65,70,75,85,55,60,68,72,80,66,70,78,58,69],
    "Previous_Score":[40,45,55,60,70,35,42,50,58,68,47,54,63,38,52],
    "Extra_Class":["No","No","Yes","Yes","Yes","No","No","Yes","Yes","Yes","No","Yes","Yes","No","Yes"],
    "Pass":[0,0,1,1,1,0,0,1,1,1,0,1,1,0,1]

}

df = pd.DataFrame(data)
print(df)
```

```
    Study_Hours  Attendance  Previous_Score Extra_Class  Pass
0             2          60              40         No     0
1             4          65              45         No     0
2             6          70              55        Yes     1
3             8          75              60        Yes     1
4            10          85              70        Yes     1
5             1          55              35         No     0
6             3          60              42         No     0
7             5          68              50        Yes     1
8             7          72              58        Yes     1
9             9          80              68        Yes     1
10            4          66              47         No     0
11            6          70              54        Yes     1
12            8          78              63        Yes     1
13            2          58              38         No     0
14            5          69              52        Yes     1
```

```
df_encoded = pd.get_dummies(df,columns=["Extra_Class"],drop_first=True)
print(df_encoded)
```

```
    Study_Hours  Attendance  Previous_Score  Pass  Extra_Class_Yes
0             2          60              40     0            False
1             4          65              45     0            False
2             6          70              55     1             True
3             8          75              60     1             True
4            10          85              70     1             True
5             1          55              35     0            False
6             3          60              42     0            False
7             5          68              50     1             True
8             7          72              58     1             True
9             9          80              68     1             True
10            4          66              47     0            False
```

```
11            6          70              54     1             True
12            8          78              63     1             True
13            2          58              38     0            False
14            5          69              52     1             True
```

```
X = df_encoded.drop("Pass",axis=1)
y = df_encoded["Pass"]
```

```
X_train,X_test,y_train,y_test = train_test_split(X,y,test_size=0.3, random_state=42)
```

```
model= RandomForestClassifier(n_estimators=100,random_state=42)
model.fit(X_train,y_train)
```

```
▾ RandomForestClassifier
▶ Parameters
```

```
y_pred = model.predict(X_test)
```

```
accuracy = accuracy_score(y_test,y_pred)
print(accuracy)
```

```
1.0
```

```
print(classification_report(y_test,y_pred))
```

```
              precision    recall  f1-score   support

           0       1.00      1.00      1.00         3
           1       1.00      1.00      1.00         2

    accuracy                           1.00         5
   macro avg       1.00      1.00      1.00         5
weighted avg       1.00      1.00      1.00         5
```

```
feature_importance = pd.DataFrame({"Feature":X.columns, "Importance":model.feature_importances_})
feature_importance.sort_values(by="Importance",ascending=False)
```

	Feature	Importance
1	Attendance	0.295918
0	Study_Hours	0.244898
2	Previous_Score	0.244898
3	Extra_Class_Yes	0.214286